

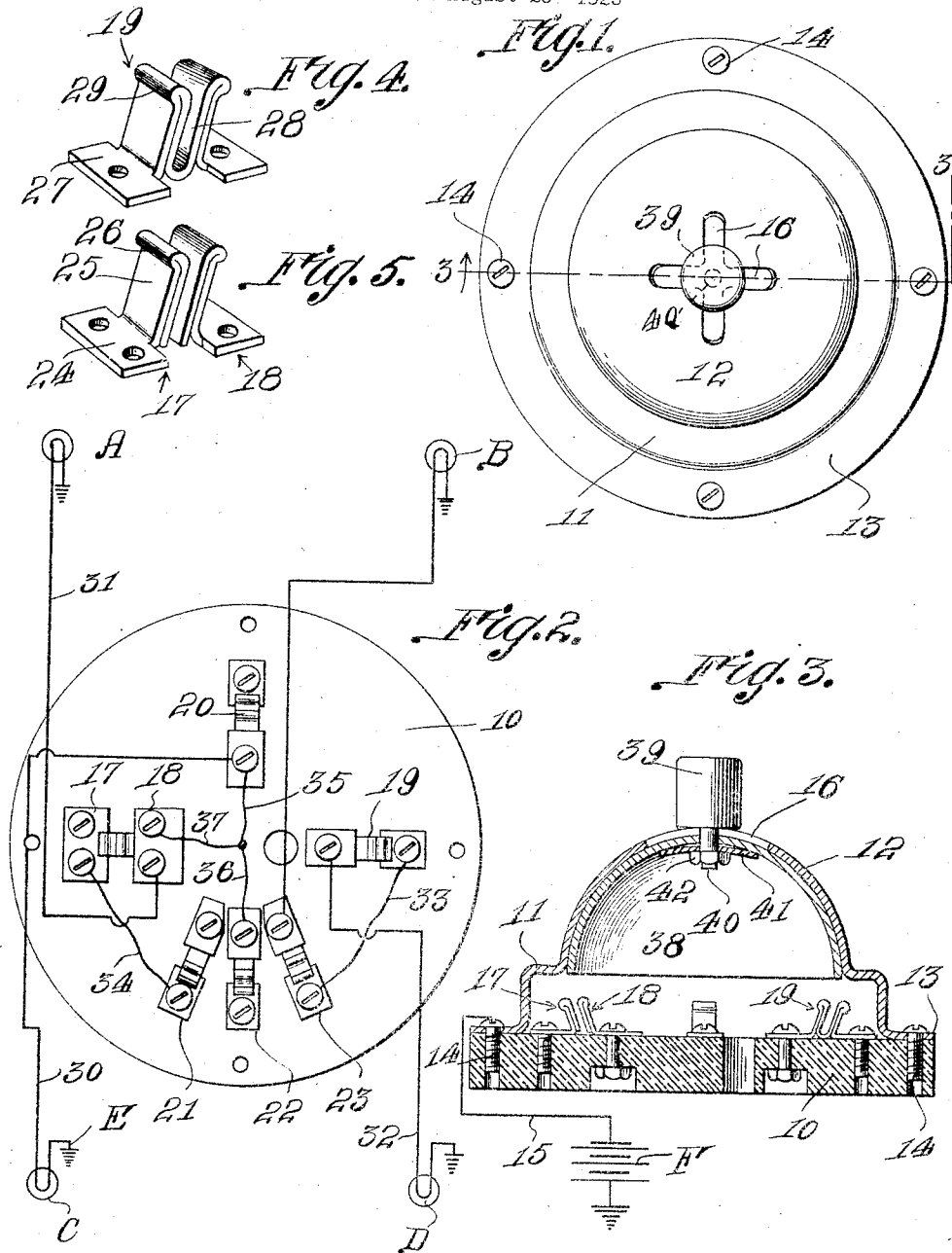
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SIGNAL LIGHT CONTROL SWITCH

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WITNESS:

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UNITED STATES PATENT OFFICE.

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SIGNAL-LIGHT-CONTROL SWITCH.

Application filed August 29, 1925. Serial No. 53,399.

This invention relates to circuit closers and has for its object the provision of a novel switch designed for use in controlling different circuits, particularly for controlling the energization of lights constituting a signal system for indicating the direction of travel of an automobile or the like.

In my co-pending application Serial No. 679,450, filed December 8, 1923, I have disclosed a signal system in which use is made of a pair of signal lamps or lights mounted at the front of the vehicle and another pair mounted at the rear, which lamps are adapted to be energized at the will of the operator and in a certain predetermined manner for indicating a contemplated turn in the direction of travel of an automobile, or an intended stop. Furthermore the signal lamps are capable of employment as parking lights.

The object of the present invention is to provide a switch involving a single control element and movable into any one of a plurality of positions for controlling the energization of the signal lights.

A more specific object is to provide a switch device in which the control member or handle has what may be called a universal movement somewhat similar to the movement of the gear shift lever on an automobile, the position of the handle indicating at a glance just which signal lamps are energized.

An additional object is to provide a device of this character which will be simple and inexpensive in manufacture, easy to apply or install, convenient to operate, efficient and durable in service, and a general improvement in the art.

With the above and other objects and advantages in view, the invention consists in the details of construction and arrangement of parts to be hereinafter more fully described and claimed, and illustrated in the accompanying drawing in which:

Figure 1 is a plan view of the device,

Figure 2 is a plan view of the base thereof showing the contacts thereon,

Figure 3 is a cross section on the line 3—3 of Figure 1,

Figure 4 is a perspective view of one type of contact, and

Figure 5 is a perspective view of a pair of contacts of a different type.

Referring more particularly to the drawings, the letters A, B, C and D represent in-

candescent lamps which are here represented as having one terminal grounded at E, the lamps being, in actual practice, of the single contact type. The letter F indicates the source of current which may be the storage battery of the car.

In carrying out the invention I provide a switch device which comprises a suitable base 10 of insulating material and of any suitable size and shape. Any preferred means may be employed for mounting this base upon any preferred location in the vehicle. Secured to and rising from the base is a shell or casing 11 which has its major portion 12 of any semispherical shape and which has its bottom open and formed with an outstanding flange 13 bearing upon and secured to the base as by screws 14 or their equivalent. The shell is connected with the source of current F as by means of a conductor 15 connected with one of the screws 14. At its top the shell is formed with four radial slots 16. While four are specified, it should be understood that a greater or lesser number may be provided depending upon the number of signal lights or other electrical devices to be controlled.

Secured upon the base 10 toward one side thereof are contact members 17 and 18 spaced apart as shown, and mounted upon the base at the opposite side of its center is a contact 19. At a point 90° from these contacts the base carries a contact 20 similar to the contact 19, while at another point are provided contacts 21, 22 and 23 arranged close together but out of contact with one another. The detailed construction of the contacts 17 and 18 is shown in Figure 5 from an inspection of which it will be seen that each is formed from a single piece of metal cut and bent to have an attaching portion 24 and an upstanding portion 25 formed as an elongated tongue bent backwardly upon itself with the bend curved outwardly as shown at 26. When the two contacts 17 and 18 are arranged in operative relation as shown the outwardly curved bent portions 16 will define a flaring entrance to the space between the contacts. All of the contacts 19, 20, 21, 22 and 23 are identical in construction and, referring to Figure 4, it will be seen that each is formed from a single piece of metal cut and bent to have attaching portions 27 connected by a U-shaped portion 28, the sides of the U being curved outwardly as shown at 29 to define a flaring entrance.

The contact 20 is connected by a conductor 30 with the lamp C, while the contact 18 is connected by a conductor 31 with the lamp A. The contact 19 is connected by a conductor 32 with the lamp D and by a conductor 33 with the contact 23. The contact 17 is connected by a conductor 34 with the contact 21, and the contacts 20, 22 and 18 are connected by conductors 35, 36 and 37.

Located within the spherical portion 12 of the shell or casing is a semispherical metallic movably mounted member 38 with which is connected a handle 39 carried by a stem 40 which passes through the slotted portion of the shell and through a spring washer 41 on the inside of the member 38, securing being effected as by a nut 42. The washer 41 will act to hold the member 38 in firm contact with the shell so as to avoid looseness while permitting the shell to be moved into any selected position as determined by the slots 16.

In the operation, it will be seen that when the handle 39 is in its intermediate position, as shown in Figures 1 and 3, the member 38 will be out of engagement with the contacts on the base. If the operator wishes to energize the lamp C, he moves the handle 39 forwardly so as to bring the edge of the member 38 into engagement with the contact 20, whereupon current will flow through the conductor 30 to the lamp C. If the operator wishes to energize the lamps A and C he moves the handle 39 to the left so that the edge of the member 38 will enter between the contacts 17 and 18, thus bridging them and feeding current through the conductor 31 to the lamp A and through the conductors 37, 35 and contact 20 and thence through the conductor 30 to the lamp C. If the operator wishes to energize the lamps B and D, he moves the handle 39 to the right to bring the member 38 into engagement with the contact 19, whereupon current will flow through the conductor 32 to the lamp D and through the conductor 33 to the contact 23 and conductor connected therewith to the lamp B. To signal a turn to the left both the lamps A and C are energized while to signal a turn to the right both lamps B and C are energized. Energization of the lamp C alone is effected only when parking so that it will be unnecessary to make use of the usual auxiliary parking lamp. If the operator wishes to energize all the lights for signalling an

intention to stop, he just pulls the handle 39 rearwardly so as to bring the member 38 into engagement with the contacts 21, 22 and 23 whereupon current will flow to all the lamps in an obvious manner.

From the foregoing description and a study of the drawings it will be apparent that I have thus provided a simply constructed, inexpensive and very easily operated control switch adapted for use for a wide variety of purposes though particularly suitable for use as a direction signalling system. It is believed that the construction, operation and advantages will be readily apparent to one skilled in the art without further explanation.

While I have shown and described a preferred embodiment of the invention, it should be understood that I reserve the right to make such changes in the form, construction and arrangement of parts as will not depart from the spirit of the invention or the scope of the subjoined claims.

Having thus described the invention, I claim:

1. In a device of the character described, a base of insulating material, a shell mounted thereon and having a partially spherical portion, a partially spherical member slidably mounted within said spherical portion of the shell and movable in a plurality of right angular directions, and a plurality of contacts mounted upon the base in position to receive and be bridged by the edge portion of said movable member, said contacts being formed to be engaged against the outer and inner surfaces of said movable member.

2. A device of the character described, a base of insulating material, a shell mounted thereon and having a partially spherical member slidably mounted within said spherical portion of the shell and movable in a plurality of right angle directions, and a plurality of contacts mounted upon the base in position to receive and be bridged by the edge portion of said movable member, said contacts being formed to be engaged against the outer and inner surfaces of said movable member, and certain of said contacts being arranged in groups and certain contacts of the groups being in electrical connection with one another.

In testimony whereof I affix my signature.

MARK E. JOHNSON.